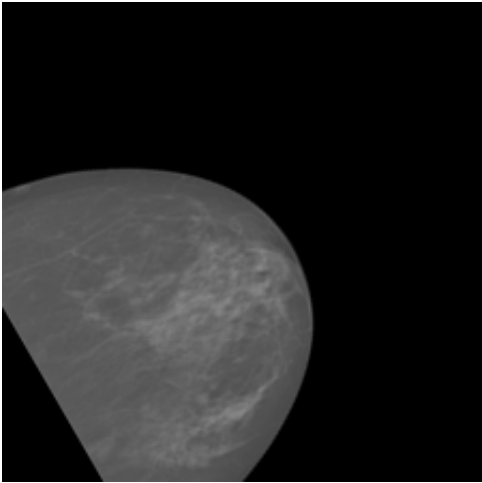


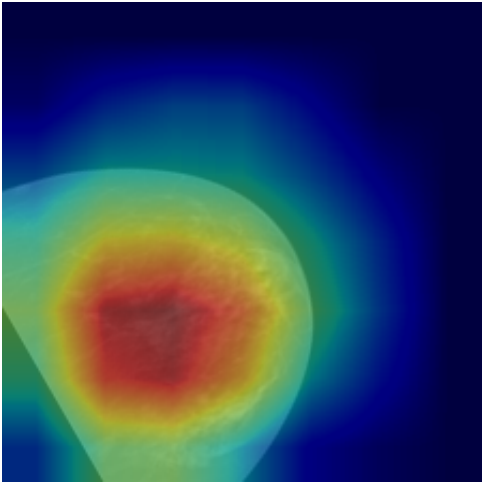
Clinical Diagnostic & RAG Report - Breast Cancer

Target Domain:	Breast Cancer	Image File:	PIL_Image
Predicted Class:	benign	Confidence:	92.7%
95% Conformal Set:	['benign', 'malignant']	Uncertainty Status:	[WARNING] Human Review Recommended

Visual Diagnostic Imaging & Grad-CAM Heatmap



Original Medical Scan



Grad-CAM Saliency Overlay (ROI)

AI Clinical Explanation (FLAN-T5 Grounded RAG)

'The image was classified as 'benign' with 92.7% confidence. Biomedical literature (PMID:35391913) indicates: The proposed method has achieved state-of-the-art results, with an accuracy of 0.9070 and the area under the receiver operating characteristic curve of 0.9459. The results demonstrate that the proposed method has the potential to be used in the clinical diagnosis of benign and malignant breast cancer.'

Faithfulness & Quantitative Text Metrics

NLI Faithfulness Score:	0.3340 / 1.0000	ROUGE-1:	0.8000
BLEU-4:	0.7636	BERTScore Similarity:	0.5456

Retrieved PubMed Literature Evidence

[1] Ultrasound Image Texture Feature Learning-Based Breast Cancer Benign and Malignant Classification. (PMID: 35391913)

'The proposed method has achieved state-of-the-art results, with an accuracy of 0.9070 and the area under the receiver operating characteristic curve of 0.9459. The results demonstrate that the proposed method has the potential to be used in the clinical diagnosis of benign and malignant breast cancer.'

[2] Diagnosis of Benign and Malignant Breast Nodules by Conventional Ultrasound in Combination with S-Detect Technology and Elastic Imaging. (PMID: 39410680)

'CONCLUSION: The combination of conventional ultrasound BI-RADS classification with elasticity score or S-Detect technology has a higher diagnostic efficacy for breast nodules, which can improve breast cancer detection and provide valuable diagnostic evidence for clinical practice. KEY WORDS: S-Detect technology, Ultrasound elastic imaging, Elasticity scoring, Elasticity strain ratio value, Breast tumour.'

[3] Ultrasensitive Ultrasound Microvessel Imaging for Characterizing Benign and Malignant Breast Tumors. (PMID: 31530420)

'To evaluate the performance of UMI, we studied 44 patients with 51 breast masses. Tumor pathology served as the gold standard: 28 malignancies and 23 benignities. UMI provided a significant improvement in depicting smaller vessels compared with conventional Doppler.'

[4] Radiomics in breast cancer classification and prediction. (PMID: 32371013)

'Breast Cancer (BC) is the common form of cancer in women. Its diagnosis and screening are usually performed through different imaging modalities such as mammography, magnetic resonance imaging and ultrasound. However, mammography and ultrasound-imaging techniques have limited sensitivity and specificity both in identifying lesions and in differentiating malign from benign lesions, especially in presence of dense breast parenchyma.'

[5] Diagnosis of Benign and Malignant Breast Nodules by Conventional Ultrasound in Combination with S-Detect Technology and Elastic Imaging. (PMID: 39410680)

'METHODOLOGY: The study included all breast nodules diagnosed using ultrasound, with patients undergoing conventional ultrasound for BI-RADS classification, elasticity score, and S-Detect examination. Benign and malignant breast nodules were classified according to the three tests and their combinations. The diagnostic sensitivity, specificity, accuracy, positive predictive value, negative predictive value, and area under curve (AUC) of those alone and combinations were calculated and compared.'